DECISION SHEET

Standard(s)- (year and edition):	**Sub clause(s):**	**Sheet n°:**
IEC 60601-1:1988 Ed.2 Am1+Am2	20.2 B-d and B-e	**DSH-428**
Subject: Insulation between an F-type applied part and enclosure	**Key words:** F-type, applied part, enclosure	**Confirmed by CTL at its 39th meeting, in Cologne**

Question:

1) Different interpretations may occur about whether to use B-d or B-e to state the correct insulation level between an F-TYPE APPLIED PART and the ENCLOSURE.

2) B-e refers to “voltages stressing the insulation”.

What is a hazardous voltage stressing an insulation?

Decision:

B-d is always applicable for a F-TYPE APPLIED PART. U equals maximum RATED SUPPLY VOLTAGE; or, 250V for INTERNALLY POWERED EQUIPMENT. BASIC INSULATION required.

In addition B-e is also applicable if there is a voltage in the F-TYPE APPLIED PART. U equals the voltage stressing the insulation in NORMAL USE including earthing of any PATIENT CONNECTION.

Explanatory notes:

B-d insulation is required for F-TYPE APPLIED PARTS.

B-e insulation is required for F-TYPE APPLIED PARTS containing internal voltages, with a reference voltage equal to the internal voltage.

The second sentence in the statement in 20.3 concerning the reference voltage between an F-TYPE APPLIED PARTand the ENCLOSURE, applies to B-d only and not to B-e.